

Clinical Decision Making and Pattern Recognition in Health Care

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August 11, 2026

1. Concept Definition

Clinical decision making and pattern recognition in health care refers to the family of computational techniques, including classification, prediction, inference, clustering, time-series anomaly detection, chain reasoning, and agentic generative AI, that surface actionable signals from complex, high-volume data. Within Treatment, Payment, and Operations (TPO) workflows, this concept applies directly to Cotiviti's core mission of payment integrity: identifying improper, fraudulent, or anomalous claims before or after payment, and doing so with a defensible, auditable rationale rather than a black-box score.

2. Relevant Trends

Classification and prediction remain the workhorses of claims risk scoring. Supervised models predict which claims are likely to be denied, delayed, or fraudulent before submission. One of the earliest healthcare-specific examples trained classifiers on Claim Adjustment Reason Codes to flag denial-prone claims with high accuracy (Saripalli et al., 2017), and predictive risk scoring is now standard in commercial payment-integrity platforms (Claritev, 2026).

Clustering groups providers or claims by behavioral similarity and often precedes anomaly detection. A claim only reads as anomalous relative to the cluster of normal peers it belongs to (du Preez et al., 2024).

Inference, tracing which billing features or contract terms are statistically associated with over or under payment, complements detection by explaining root causes rather than just flagging symptoms (SRM Tech, 2025).

Time-series anomaly detection is shifting from static, rule-based claims edits toward reconstruction-error models such as autoencoders, which better capture subtle, non-linear billing irregularities than distance-based methods alone (Bairy et al., 2024). This mirrors work I completed as a graduate researcher at the University of Florida, where I built a PCA-based autoencoder for unsupervised anomaly detection on time-series network data, achieving an F1 score of 0.868 (precision 0.766, recall 1.000) by switching from mean-based to max-based reconstruction-error aggregation, a finding accepted for publication at IEEE NAPS 2026 (Mittal et al., 2026). The same principle applies to claims: the anomalies that matter most are often the worst single deviation in a claim, not the average deviation across a provider's history, which mean-based scoring tends to dilute.

Chain reasoning is displacing single-shot classification in clinical and administrative decision support. Recent clinical LLM-agent benchmarks show that decomposing a decision into explicit, step-by-step reasoning, rather than one-shot judgment, substantially improves performance on sequential, multi-step cases and is now a standard architectural pattern for agent evaluation (Schmidgall et al., 2024).

Agentic generative AI extends chain reasoning with autonomous tool use and retrieval. Retrieval-augmented agents that can search contracts, policies, and prior claims before responding are increasingly proposed for healthcare claim adjudication and audit workflows, though current reviews note most agentic healthcare systems remain exploratory and lack robust real-world validation (Collaco et al., 2026).

Across all of these techniques, explainability has become a deployment requirement rather than an optional feature. Reviews of agentic AI in health care consistently identify the black box problem, an unauditable reasoning chain, as the primary barrier to clinical and regulatory trust (Collaco et al., 2026), a concern deep learning anomaly-detection surveys echo specifically for fraud domains (Chalapathy & Chawla, 2019).

3. Opportunities

Combining these techniques, clustering to define normal, anomaly detection to flag deviation, inference to explain the driver, and an agentic layer to retrieve evidence and reason through it step by step, lets Cotiviti catch fraud, waste, and abuse patterns that single-technique models miss, particularly emerging schemes with no historical label (Bairy et al., 2024). An agentic pipeline that pairs anomaly scoring with retrieval and chain-reasoned explanation can compress reviewer time by surfacing not just what looks anomalous but why, citing the specific contract clause or billing pattern behind the flag. This is a natural extension of Cotiviti's existing payment integrity stack and would strengthen both prepay accuracy and postpay recovery, while giving reviewers an audit trail they can defend to providers and regulators.

4. Threats and Risks

False positives at scale risk provider abrasion and rising appeals volume. Systematic reviews note this remains one of the field's persistent, unresolved challenges given class imbalance and inconsistent benchmark datasets across the industry (du Preez et al., 2024).

Agentic and chain-reasoning systems introduce additional exposure beyond a single classifier: multi-agent coordination failures, model drift as billing patterns evolve, and regulatory risk if a reasoning trace cannot be independently verified during an audit (Collaco et al., 2026). Data privacy and HIPAA constraints further limit how much raw claims data can be used for continuous model retraining without careful de-identification and governance, and any pilot needs a clear ownership model for when the agent's rationale and a human reviewer's judgment disagree.

5. Strategic Recommendation

Cotiviti should pilot a narrow-scope agentic system on one claims category, combining all seven techniques in a single pipeline: clustering to define peer-group norms, a classifier for prior risk scoring, an autoencoder-based anomaly score (validated the way max-aggregation improved sensitivity in the NAPS 2026 work), an inference step identifying the likely driver, and a chain-reasoning, retrieval-augmented agent that explains and cites the specific policy or historical pattern behind each flag. Piloting on a single, bounded category keeps the audit surface small, lets reviewers validate the reasoning trace against real cases before wider rollout, and builds the internal

reasoning trace standard that emerging XAI-in-agentic-AI literature identifies as the deciding factor for adoption (Collaco et al., 2026).

6. References

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